Do Now · retrieval

Lesson 126 · Addressing Modes: Immediate, Direct, Indirect and Indexed

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which LMC instruction reads a mailbox value into the accumulator?

Q2

In a BST, which side holds a smaller value?

Q3

How many bits are in three bytes?

Q4

What is a function parameter?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Immediate versus direct — LOAD #12 uses literal 12; LOAD 12 uses contents of memory[12].

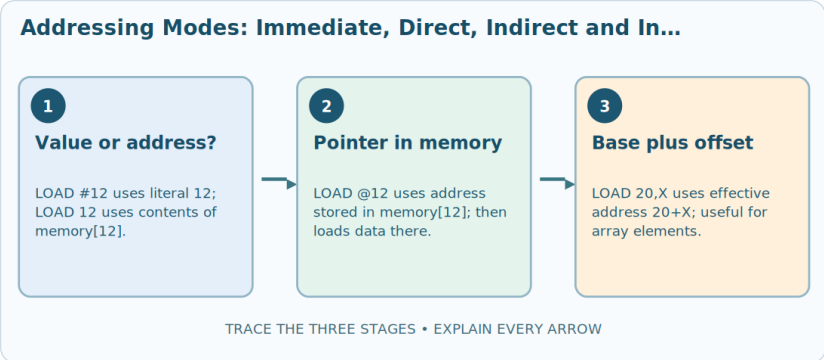
Core idea 2

Indirect addressing — LOAD @12 uses address stored in memory[12]; then loads data there.

Core idea 3

Indexed addressing — LOAD 20,X uses effective address 20+X; useful for array elements.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

immediate — Operand is a literal value.

direct — Operand gives the memory location holding a value.

indirect — Operand points to a location holding the effective address.

indexed — Base address plus index gives effective address.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

With memory[12]=40 and memory[40]=7, compare LOAD #12, LOAD 12 and LOAD @12.

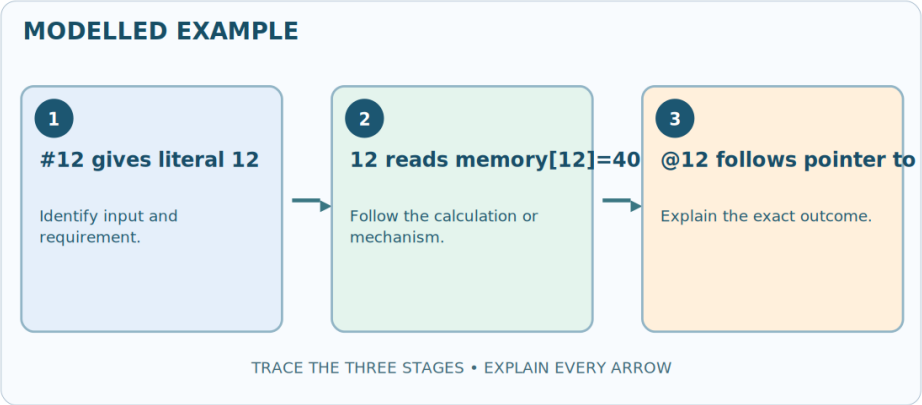
Follow the worked method

1. #12 gives literal 12 — Predict the first step from the prompt.

2. 12 reads memory[12]=40 — Explain the intermediate result or decision.

3. @12 follows pointer to 7 — Using illustrative assembler conventions: immediate gives 12; direct reads 40; indirect follows memory[12]=40 then loads memory[40]=7.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Base 80, index 3
Predict the first step from the prompt.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define immediate and direct addressing with one example each. [6]

Question 2

Memory[15]=29 and memory[29]=6. Give values loaded by direct LOAD 15 and indirect LOAD @15. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain indexed addressing for an array at base 50 with index 4 and discuss why this is not simply LMC LDA notation. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In direct addressing, the operand always means the literal value to be used.

Use key vocabulary from memory

1. Explain immediate in your own words.

2. Explain direct in your own words.

3. Explain indirect in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.